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ABOUT ME

Software Engineer specializing in cloud-native systems and system reliability. Experienced in developing **microservices (Go, JavaScript, Python)** and managing the full application lifecycle using **Docker** and **Kubernetes** across AWS and Azure. implementing **CI/CD pipelines** and enhancing system traceability through observability tools like **OpenTelemetry, Prometheus**.

OPENSOURCE CONTRIBUTION

ECLIPSE KUKSA DATABROKER (RUST)

Contributed to the [Eclipse kuksa-databroker](#) by integrating **OpenTelemetry for distributed tracing**. This improved system observability, **enabled trace analysis across microservices**, and helped identify latency bottlenecks in automotive data communication.

EXPERIENCE

ROBERT BOSCH MOBILITY

OCT 2025 – MAR 2026

MASTER THESIS: FAULT INJECTION MECHANISMS IN VIRTUALIZED ENVIRONMENTS AND ITS

OBSERVABILITY | Go, Docker, Kubernetes, eBPF, Prometheus, Thanos, AWS (EC2, S3 bucket, Lambda functions), ReactJS

- Deployed cross-platform **observability exporters** (Node, Process, Windows, custom eBPF) as Kubernetes Daemon Sets and utilized `kubernetes_sd_configs` for dynamic discovery and label-based controls to manage resource overhead.
- Developed a **full-stack observability suite** featuring interactive metrics visualization dashboard with log aggregation and fault Injection.
- Engineered a fault injection framework to introduce controlled resource contention and analyzed its impact on vECU simulations using the observability suite.
- Implementing USE Method to ease tracking the performance related issues.

ROBERT BOSCH CORPORATE RESEARCH

FEB 2025 – JULY 2025

INTERN | RUST, Go, Docker, OpenTelemetry, Grafana, WebRTC, Microsoft Azure

Low-latency communication strategies to enable Vehicle-to-Cloud (V2C) teleoperation. The goal is to reduce end-to-end latency in teleoperation of vehicles by evaluating and integrating WebRTC-based communication for real-time streaming and signaling.

- Evaluating **Microsoft Azure** infrastructure components including:
 - **Proximity placement groups** to ensure physical co-location of compute resources
 - Evaluating the effect of **VNet** Peering for Low-Latency Interconnects
 - Architectural Considerations for Cloud **Scalability** and **Fault Tolerance**
- Analysis of Peer-to-Peer Network Paths Across MNOs, ISPs, and Cloud Infrastructure in Wired and Wireless scenarios.

ROBERT BOSCH CORPORATE RESEARCH

FEB 2024 – NOV 2024

WORK STUDENT | RUST, Go, Docker, OpenTelemetry, Jaeger, Grafana, Prometheus, eBPF, ROS2, gRPC

Black box testing of [kuksa-databroker](#) using `ghz` tool. Modified the [ghz tool](#) written in **Go language** to capture custom metrics helpful in black box testing to obtain end-to-end latency of **kuksa-databroker**.

White box testing of [kuksa-databroker](#) using OpenTelemetry in **rust** programming language.

- Accomplished comprehensive observability of the [kuksa-databroker](#) by implementing an **OpenTelemetry** pipeline to track function spans, propagate trace contexts, and optimize data collection, which led to a unified view of application performance for analysis in Jaeger.
- Integrated **Prometheus** for metrics collection to monitor service-level performance and expose custom application metrics alongside tracing data.

This observability setup helps in debugging, monitoring performance, identifying latency bottlenecks, and diagnosing failures by giving a comprehensive view of the application's behavior and interactions.

Multi-Platform Framework for Vehicular Functions

Developed a software framework to deploy a single codebase for vehicular functions across simulation, model, and full-scale ROS-based vehicle (FlexCAR), significantly reducing development time.

Developed a VSS-to-ROS communication bridge using a Raspberry Pi to enable non-invasive teleoperation and GStreamer video feedback from a remote-control center.

UNIVERSITY OF STUTTGART

MAY 2023 – JAN 2024

HIWI | Python, ROS2, MATLAB

Fault injection and detection in Franka Emika panda arm by making time series analysis of input signals sent to the arm. And bring the arm to halt when a faulty input is detected. Signal processing was done in **MATLAB Simulink** alongside **ROS** package in MATLAB Simulink for communication with robotic arm.

SofDCar Hackathon – Technical Contributor & Organizer: Contributed to the organization of the SofDCar Hackathon Challenge by coordinating technical resources and preparing onboarding materials. Developed and maintained startup scripts and boilerplate code to streamline project initialization.

ENGINECAL TECHNOLOGY PRIVATE LIMITED, BENGALURU

FEB 2022 – FEB 2023

SOFTWARE ENGINEER | JavaScript, Python, MEAN stack, AWS, Microsoft Azure, Docker, Kubernetes

Worked as a Product development Engineer mainly focused on DevOps and Full-stack development.

Developed and maintained a microservices architecture for vehicle data processing with **CI/CD**, deployed on **Azure Kubernetes Service (AKS)** and **AWS Elastic Kubernetes Service (EKS)**.

ELLIPSONIC INDIA SOLUTIONS PVT LTD, BENGALURU

APR 2021 - JAN 2022

SOFTWARE ENGINEER | JavaScript, Python, AWS, MERN stack

Developed and maintained full-stack web applications using the MERN stack (MongoDB, Express.js, React, Node.js).

Managed cloud infrastructure with **AWS EC2**, **AWS Lambda**, **AWS CloudFront**.

Designed a web application to control AWS instances via **AWS CLI** with real-time status updates of EC2 instances. Used **AWS CloudWatch** and **AWS Lambda** for event-driven automation on AWS infrastructure.

Contributed to building a virtual platform for the **Ministry of Health, Singapore**. The platform included the functionality of sending email which made use of **Amazon Simple Email Service (SES)**, **Amazon CloudWatch**, **AWS Lambda**.

Performed **data engineering tasks** in Microsoft Azure, working with **Azure Synapse Analytics** to implement **ETL pipelines** using:

- **Azure Data Factory** for orchestrating data workflows,
- **Azure Data Lake Storage** for scalable storage of structured and unstructured data,
- **Synapse SQL & Spark pools** for data transformation and analysis of large datasets in preparation for analytics workflows.

EDUCATION

UNIVERSITY OF STUTTGART

GPA 1.7

M.SC ELECTRICAL ENGINEERING – SMART SYSTEMS

Relevant Coursework: Service Management and Cloud Computing, and Evaluation, Advanced Mathematics for Signal and Information Processing, Deep Learning, Detection and Pattern Recognition, Hardware Platforms and Programming of Embedded Systems, Modeling and Analysis of Automation Systems, Industrial Automation Systems.

BANGALORE INSTITUTE OF TECHNOLOGY

GPA 1.5

B.E ELECTRONICS ANS COMMUNICATIONS

SKILLS

Languages: Python, JavaScript, Go, Rust

Technologies & Tools: Docker, Kubernetes, eBPF, AWS, Azure Microsoft, OpenTelemetry, Prometheus, SQL, NoSQL, MERN Stack, MEAN Stack, gRPC, ROS.

LANGUAGE PROFICIENCY

English: C2, **German:** A2 (Learning)

PERSONAL PROJECTS

[Observability-framework](#), [VSS-ROS-Bridge](#), [terraform-ansible-highavailability-cluster](#), [fort-k8s](#), [path-finder \(wasm\)](#)